

OPERATING SYSTEMS

WEEK- 5 Practical

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Section no:- 9

5.Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Program:-



```
GNU nano 8.7.1 WEEK-5_P.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>
int main() {
int fd[2];
pipe(fd);
if (fork() == 0) {
dup2(fd[1], STDOUT_FILENO);
close(fd[0]);
close(fd[1]);
execlp("ls", "ls", "-l", NULL);
perror("exec ls failed");
exit(1);
}
if (fork() == 0) {
dup2(fd[0], STDIN_FILENO);
close(fd[0]);
close(fd[1]);
execlp("grep", "grep", ".c", NULL);
perror("exec grep failed");
exit(1);
}
```

```

}
if (fork() == 0) {
dup2(fd[0], STDIN_FILENO);
close(fd[0]);
close(fd[1]);
execvp("grep", "grep", ".c", NULL);
perror("exec grep failed");
exit(1);
}
close(fd[0]);
close(fd[1]);
wait(NULL);
wait(NULL);
return 0;
}

```

Output:-

```

priyadarshini@DESKTOP-P0IK742:~$ gcc WEEK-5_P.c
priyadarshini@DESKTOP-P0IK742:~$ ./a.out
-rw-r--r-- 1 priyadarshini priyadarshini 506 Jul 28 05:32 Arithmetic.c
-rw-r--r-- 1 priyadarshini priyadarshini 156 Aug 4 14:14 Excel.c
-rw-r--r-- 1 priyadarshini priyadarshini 0 Aug 3 08:20 Exce2.c
-rw-r--r-- 1 priyadarshini priyadarshini 0 Aug 4 14:12 Exec1.c
-rw-r--r-- 1 priyadarshini priyadarshini 242 Sep 10 04:50 Exec2.c
-rw-r--r-- 1 priyadarshini priyadarshini 0 Aug 3 06:56 Execl.c
-rw-r--r-- 1 priyadarshini priyadarshini 392 Sep 1 04:40 FCFS.c
-rw-r--r-- 1 priyadarshini priyadarshini 172 Jul 30 08:03 Process1.c
-rw-r--r-- 1 priyadarshini priyadarshini 215 Aug 3 02:28 Process2.c
-rw-r--r-- 1 priyadarshini priyadarshini 154 Aug 1 09:19 Process2.c.save
-rw-r--r-- 1 priyadarshini priyadarshini 184 Aug 3 02:30 Process3.c
-rw-r--r-- 1 priyadarshini priyadarshini 550 Aug 3 02:38 Process4.c
-rw-r--r-- 1 priyadarshini priyadarshini 550 Aug 1 10:17 Process4.c.save
-rw-r--r-- 1 priyadarshini priyadarshini 617 Sep 1 05:30 RR.c
-rw-r--r-- 1 priyadarshini priyadarshini 502 Sep 1 05:18 SJF.c
-rw-r--r-- 1 priyadarshini priyadarshini 185 Sep 10 04:54 Sys1.c
-rw-r--r-- 1 priyadarshini priyadarshini 324 Sep 10 04:57 Sys2.c
-rw-r--r-- 1 priyadarshini priyadarshini 295 Sep 10 04:53 Sys3.c
-rw-r--r-- 1 priyadarshini priyadarshini 1 Aug 25 05:42 Txt1.c
-rw-r--r-- 1 priyadarshini priyadarshini 352 Sep 10 04:53 WEEK-1_P.c
-rw-r--r-- 1 priyadarshini priyadarshini 434 Sep 10 05:18 WEEK-2_P.c
-rw-r--r-- 1 priyadarshini priyadarshini 426 Sep 10 04:56 WEEK-3_P.c
-rw-r--r-- 1 priyadarshini priyadarshini 515 Sep 10 04:58 WEEK-4_P.c
-rw-r--r-- 1 priyadarshini priyadarshini 472 Sep 10 05:46 WEEK-5_P.c
-rw-r--r-- 1 priyadarshini priyadarshini 1 Aug 11 05:53 Z01.c
-rw-r--r-- 1 priyadarshini priyadarshini 145 Jul 28 04:41 addition.c
-rw-r--r-- 1 priyadarshini priyadarshini 244 Aug 25 05:52 open.c
-rw-r--r-- 1 priyadarshini priyadarshini 277 Aug 25 06:07 read.c

```